Sheet 5

Time covariance and sector states

1. For $H = \mathrm{diag}(0, \Delta, 2\Delta)$, $\Delta > 0$, compute the return probability of $(|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$. Replace U_t by $e^{ict}U_t$ and find the new Hamiltonian and its domain. Which part of the probability changes?
2. **Wrapped energy shifts.** On $L^2(0, L)$ put $H = \alpha Q$, $T = -P_\theta/\alpha$, $\alpha > 0$, with $\psi(x + L) = e^{i\theta}\psi(x)$ for the twisted periodic extension. For $0 < s < \alpha L$ compute $W_s = e^{isT/\hbar}$ from the momentum eigenfunctions, then compute $W_s H W_s^{-1}$ as a multiplication operator. Locate the interval where it differs from $H - s\mathbb{1}$. Which covariance condition fails despite $[T, H]\psi = i\hbar\psi$ on compactly supported smooth vectors?

   > **First hint**
   >
   > Apply the exponential to one twisted momentum eigenfunction. Compare the resulting phase with evaluating that eigenfunction at a translated argument. Keep the twisted extension when the argument leaves the interval.

   > **Further hint**
   >
   > Split the interval according to whether the translated argument wraps through an endpoint. The twist phases cancel in conjugating the multiplication operator, but its coordinate multiplier changes by one interval length on the wrapped piece. Compare this piece with the proposed energy shift.

3. Let u be a unit vector in the integer sector and v one in the half-integer sector, and put $\psi = \frac{1}{2}u + i\frac{\sqrt{3}}{2}v$. Find its unique block-diagonal representative on the two-block algebra, its expectation of $2\mathbb{1}_{\mathrm{int}} \oplus \mathbb{1}_{\mathrm{half}}$, and its restricted purity status. Construct the phase-matched cross-sector witness and compute its mean in the ray and block mixture. Is that witness an admitted observable?